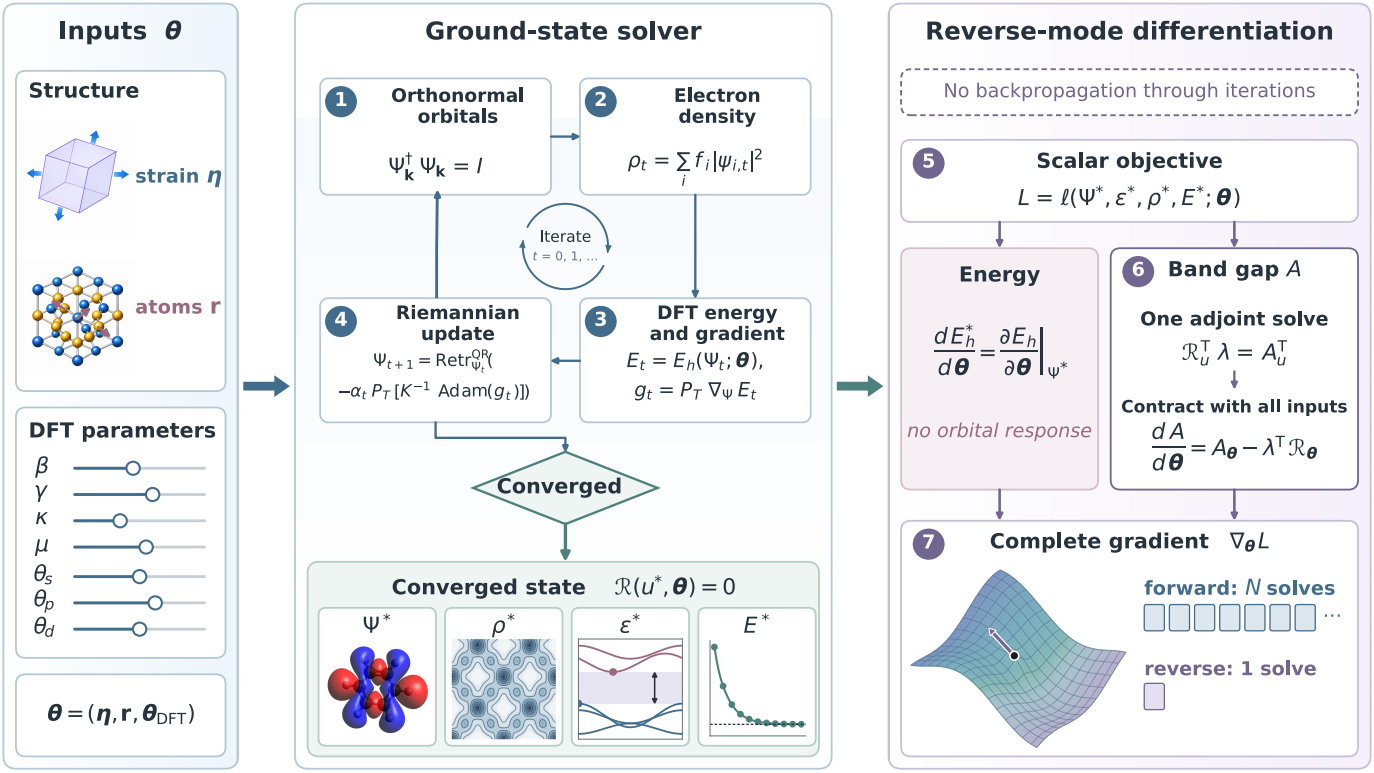

Inputs θ
Structure
strain η
atoms r
DFT parameters
β
γ
κ
μ
θs
θp
θd
θ = (η, r, θDFT)
Ground-state solver
1 Orthonormal orbitals
Ψk† Ψk = I
2 Electron density
ρt = Σi fi |ψi,t|²
Iterate
t = 0, 1, ...
4 Riemannian update
Ψt+1 = RetrΨtQR(
−αt PT [K⁻¹ Adam(gt)])
3 DFT energy and gradient
Et = Eh(Ψt; θ),
gt = PT ∇Ψ Et
Converged
Converged state R(u*, θ) = 0
Ψ*
ρ*
ε*
E*
Reverse-mode differentiation
No backpropagation through iterations
5 Scalar objective
L = ℓ(Ψ*, ε*, ρ*, E*; θ)
Energy
dE*h/dθ = ∂Eh/∂θ |Ψ*
no orbital response
6 Band gap A
One adjoint solve
Rᵘᵀ λ = Aᵘᵀ
Contract with all inputs
dA/dθ = Aθ − λᵀ Rθ
7 Complete gradient ∇θ L
forward: N solves
reverse: 1 solve
ground-state iteration
converged-state output
reverse-mode differentiation